

# LAB 6

## Polynomial Approximation

### Assignment 1:

Sample numerical.

Q. Suppose we have xdata and ydata points observed in any experiment as follows –

|     |   |   |   |   |   |    |    |    |    |    |
|-----|---|---|---|---|---|----|----|----|----|----|
| $x$ | 0 | 2 | 4 | 6 | 9 | 11 | 12 | 15 | 17 | 19 |
| $y$ | 5 | 6 | 7 | 6 | 9 | 8  | 8  | 10 | 12 | 12 |

Using hand calculations, show the linear regression for the above data also using curve fitting toolbox in MATLAB, evaluate the goodness of fits with linear and polynomial fit with 2<sup>nd</sup> degree.

## Hand Calculation:

| $x$         | $y$ | $(y - \bar{y})^2$ | $(y - a_0 - a_1x - a_2x^2)^2$ |
|-------------|-----|-------------------|-------------------------------|
| 0           | 5   | 18.49             |                               |
| 2           | 6   | 5.29              |                               |
| 4           | 7   | 1.69              |                               |
| 6           | 6   | 5.29              |                               |
| 9           | 9   | 0.49              |                               |
| 11          | 8   | 0.09              |                               |
| 12          | 8   | 0.09              |                               |
| 15          | 10  | 2.89              |                               |
| 17          | 12  | 13.69             |                               |
| 19          | 12  | 13.69             |                               |
| $\Sigma$ 95 | 83  |                   |                               |

|                 |                      |                         |
|-----------------|----------------------|-------------------------|
| $n = 2$         | $\Sigma x = 95$      | $\Sigma x^4 = 307973$   |
| $\bar{x} = 9.5$ | $\Sigma y = 83$      | $\Sigma xy = 923$       |
| $\bar{y} = 8.3$ | $\Sigma x^2 = 1277$  | $\Sigma x^2y = 13251.8$ |
|                 | $\Sigma x^3 = 19223$ |                         |

$$\begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{Bmatrix} \Sigma y \\ \Sigma xy \\ \Sigma x^2y \end{Bmatrix}$$
  

$$\begin{bmatrix} 10 & 95 & 1277 \\ 95 & 1277 & 19223 \\ 1277 & 19223 & 307973 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{Bmatrix} 83 \\ 923 \\ 13251.8 \end{Bmatrix}$$
  

$$a = \begin{matrix} 5.4044 \\ 0.1718 \\ 0.0099 \end{matrix}$$
  

$$\therefore y = 5.4044 + 0.1718x + 0.0099x^2$$

Figure 1 Hand calculation using Least square method

## Matlab Code

```
/MATLAB Drive/EXp6.m

1 x=[0,2,4,6,9,11,12,15,17,19];
2 y=[5,6,7,6,9,8,8,10,12,12];
3 cftool;
4
```

Figure 2 Matlab code for finding curve using cftool

## Result

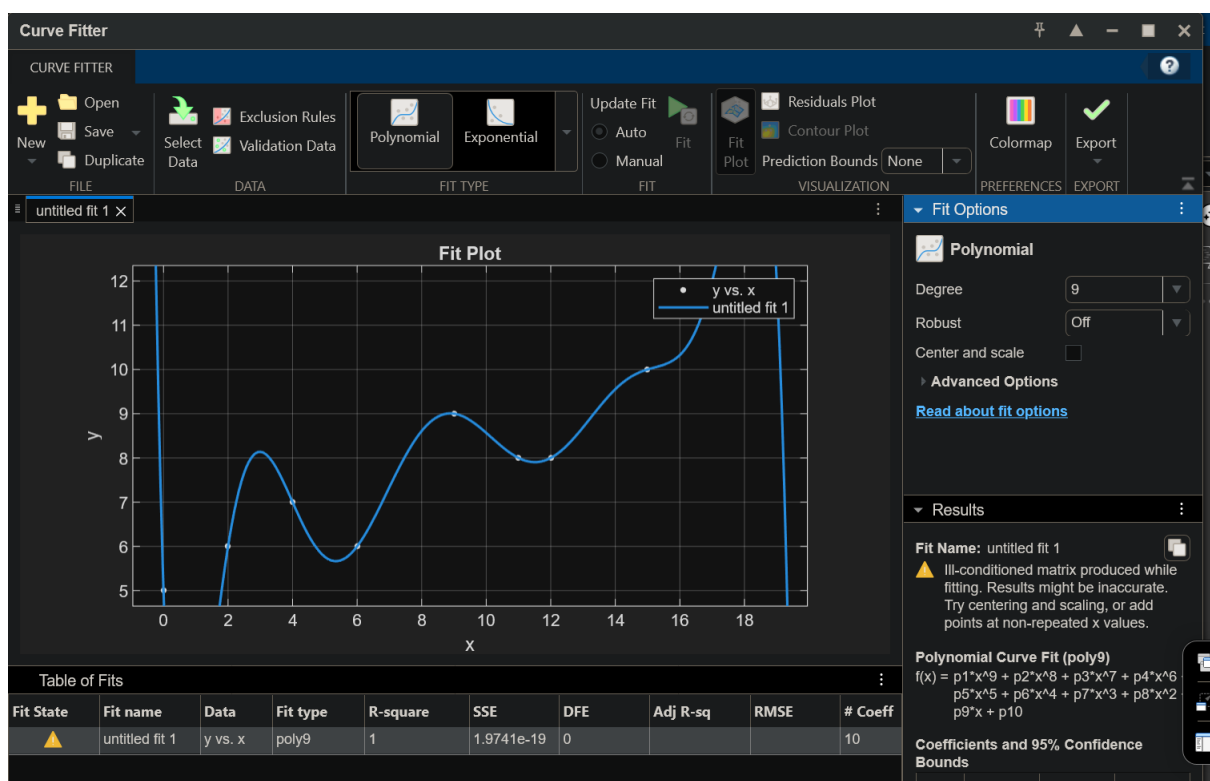


Figure 3 CF tool solution using 9 degree polynomial

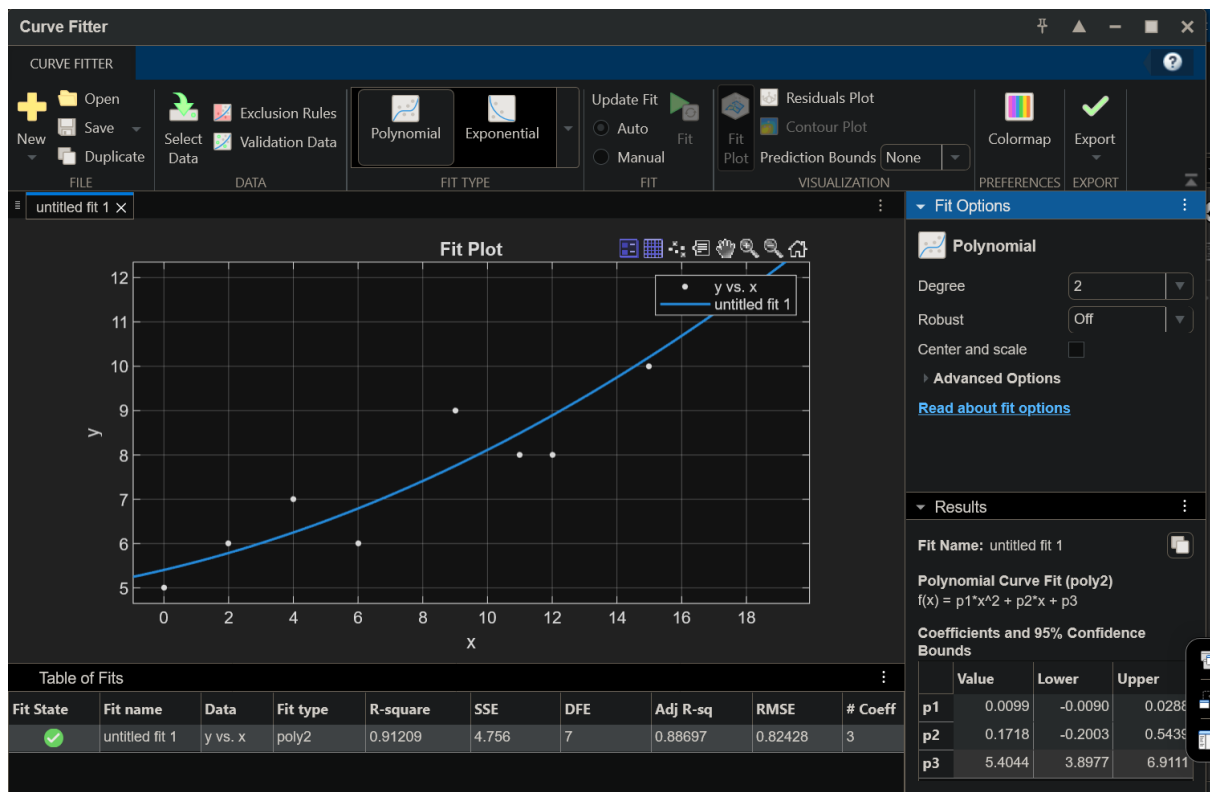


Figure 4 CF tool solution using the 2nd degree polynomial